LEC. 3.5: MACROTRANSPORT

Learning goals:

- To know about the types of macrotransport
- To understand the celular/molecular processes involved in macrotransport
- To learn about clinical scenarios where macrotransport is relevant in humans

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MICROTRANSPORT

- **No membrane remodeling**

Structure remains unchanged

- **Solute scope**

Small molecules, water
ions
gases

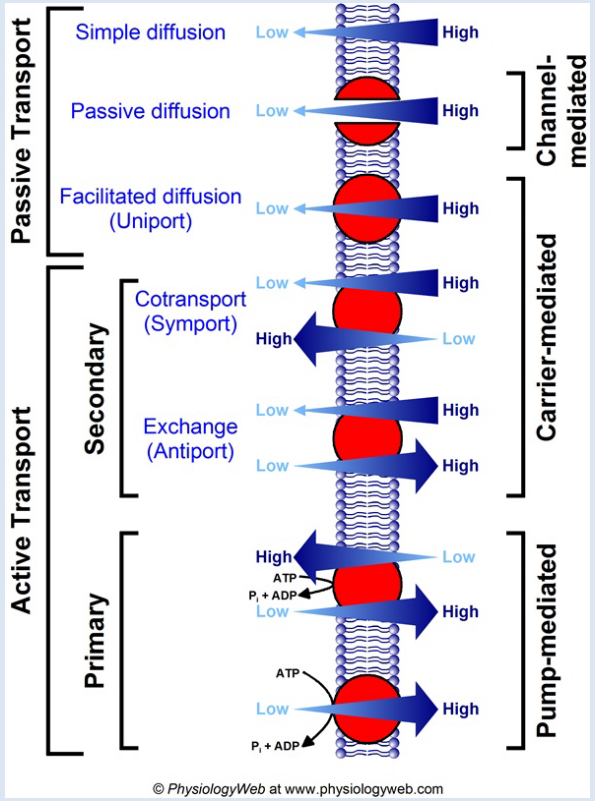
- **Proteins**

None (simple difussion)

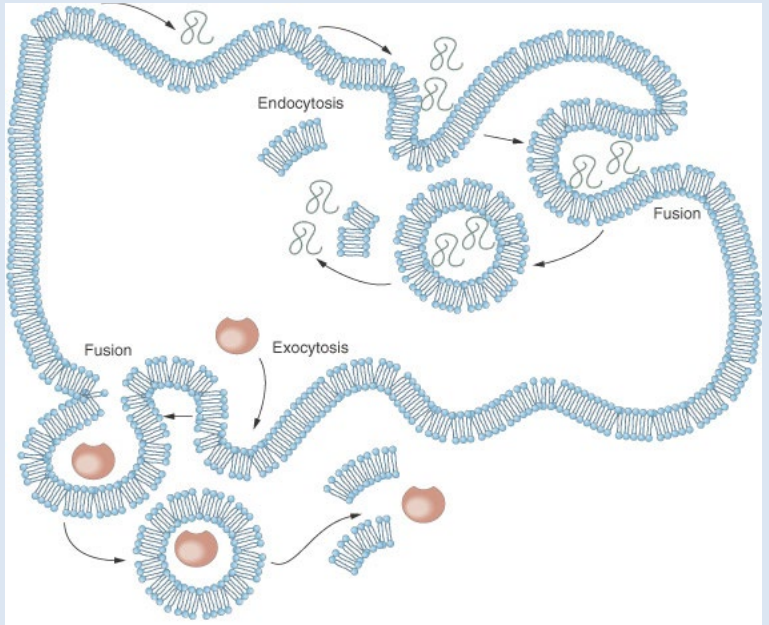
Channels

Carriers

Pumps



MACROTRANSPORT



- **Membrane remodeling**

Dynamic shape alterations

- **Solute scope**

Liquid and nutrients
Soluble ligands
Hazardous/waste particles
Micro-organisms

- **Active cytoskeletal**

& energy involvement

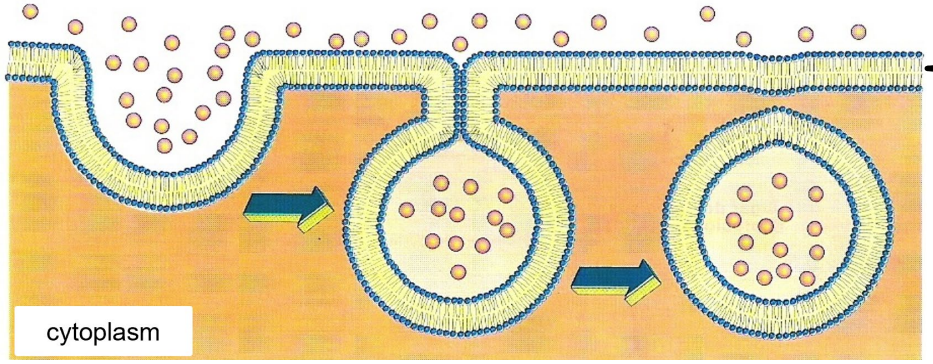
MACROTRANSPORT

Transportation of molecules at a large scale → modifications in the plasma membrane

Active energy requirement	Cytoskeletal reorganization
ATP/GTP	Actin cytoskeleton

TYPES OF MACROTRANSPORT

ENDOCYTOSIS

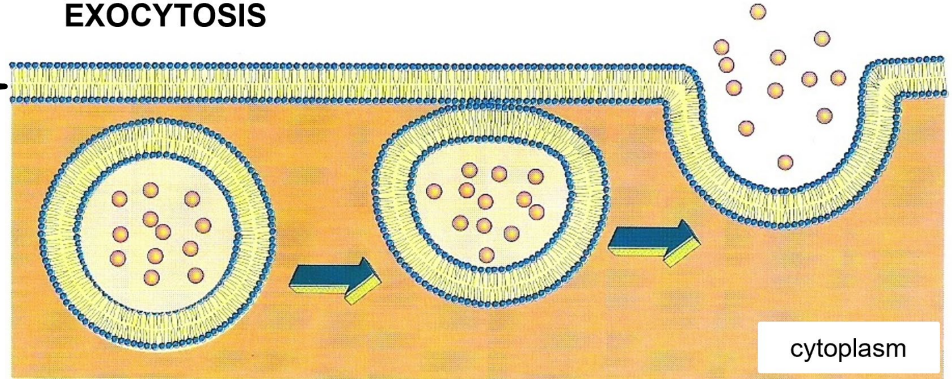


- Incorporation of external molecules/particles
- Membrane budding (e.g., to form vesicles)

ENDOCYTIC-EXOCYTIC CYCLE

- Intracellular vesicular trafficking
- Balance membrane surface area and organelle integrity

EXOCYTOSIS



- Release of cell components
- Outward vesicle fusion

1. ENDOCYTOSIS

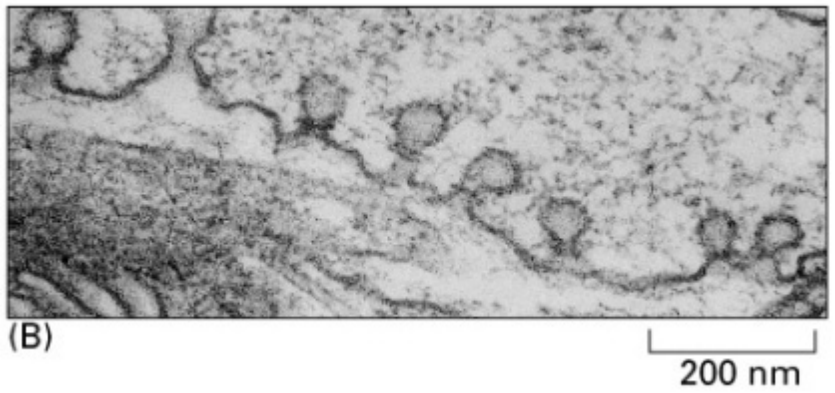
PINOCYTOSIS (“cellular drinking”)

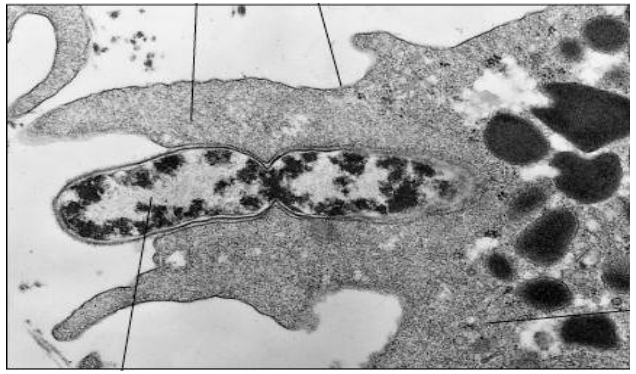
Clear content
Liquid and small dissolved particles
Vesicles ≤ 0.1 µm diameter
Almost all cells constitutively

TRANSCYTOSIS: special type of pinocytosis
(e.g., blood and intestinal cells)

PHAGOCYTOSIS (“cellular eating”)

Electron dense content
Swallowing of large particles and microorganisms
Vacuoles ≥ 0.25 µm
Mammals and higher organisms
Specialized cells (phagocytic cells)
Protozoa: regular way of feeding



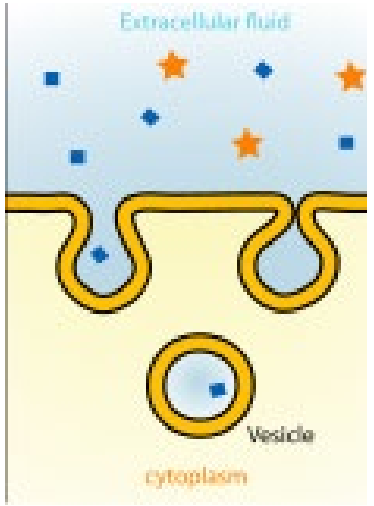


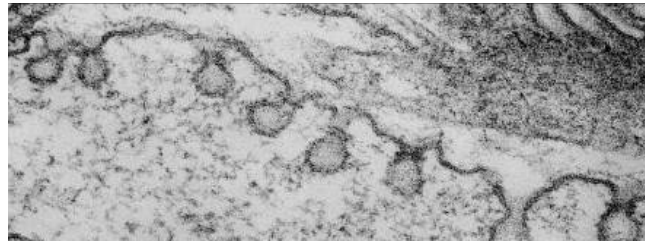
PINOCYTOSIS

TYPES OF PINOCYTOSIS

Non receptor-mediated pinocytosis
(or fluid-pase endocytosis)

- Continuous and non-selective
- Caveolae (**0.05-0.1** µM diameter)
- Caveolins (form caveolae) and
  cavins (control caveola formation/maintenance)



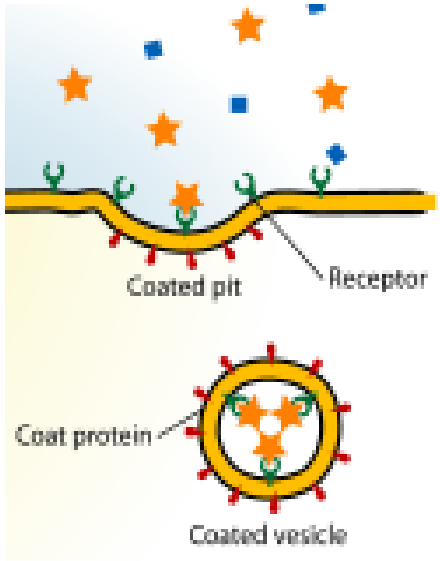


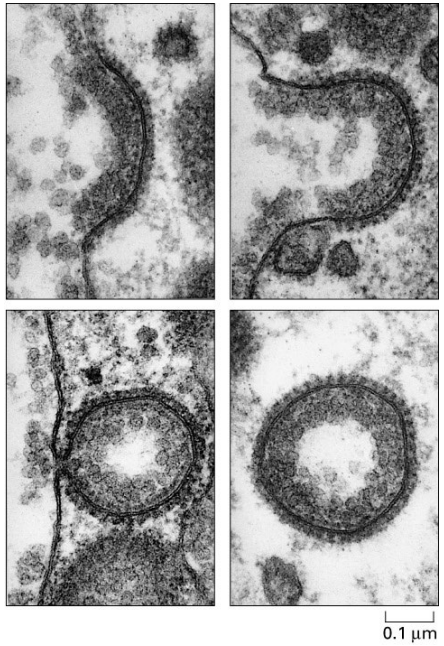
Macropinocytosis

- Continuous and non-selective
- Lamelipodia (0.5-5 µM)
- Cytoskeleton changes (actin)

Receptor-mediated pinocytosis (RME)

- Specific (selective) and highly efficient
- Electron dense coating material (**0.1** µM)





RECEPTOR-MEDIATED PINOCYTOSIS (RME)

A. VESICLE FORMATION

1. **Receptor** binding by **ligand**

2. Binding of intracellular prots to receptor: **adaptins**

3. Assembly of **clathrin** to adaptins

Coating: **adaptins** + **clathrins**

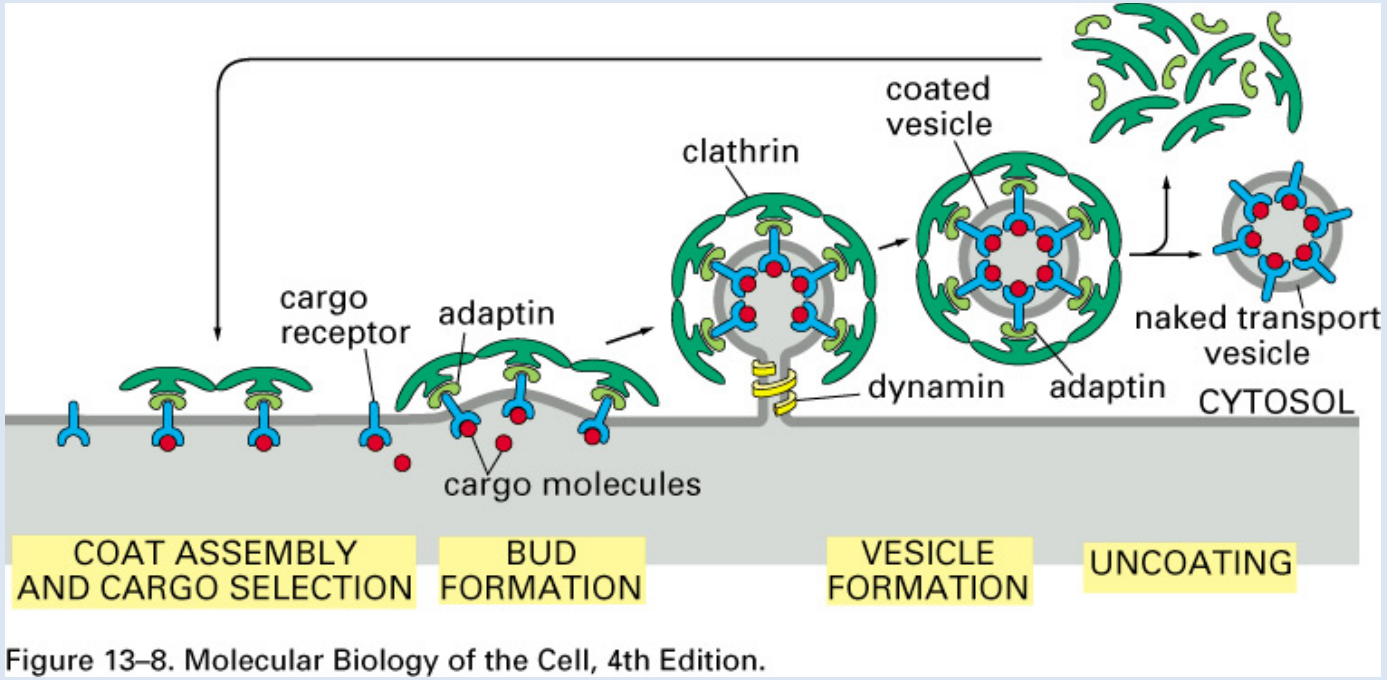


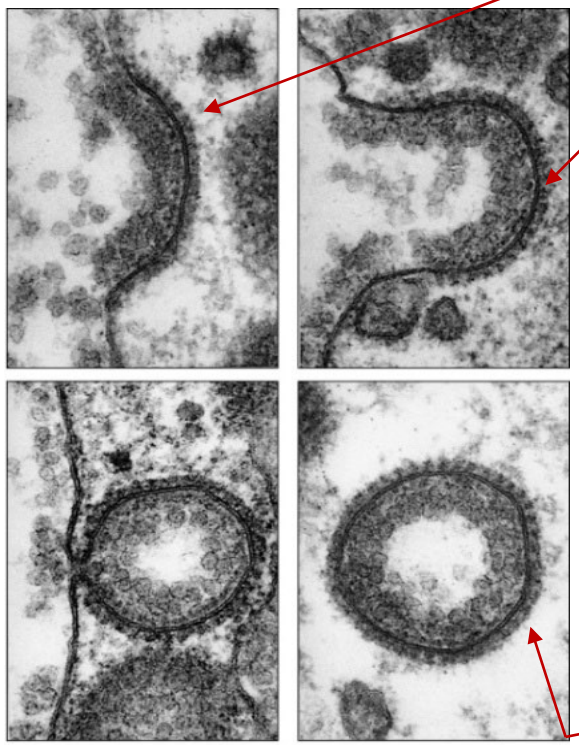
Figure 13–8. Molecular Biology of the Cell, 4th Edition.

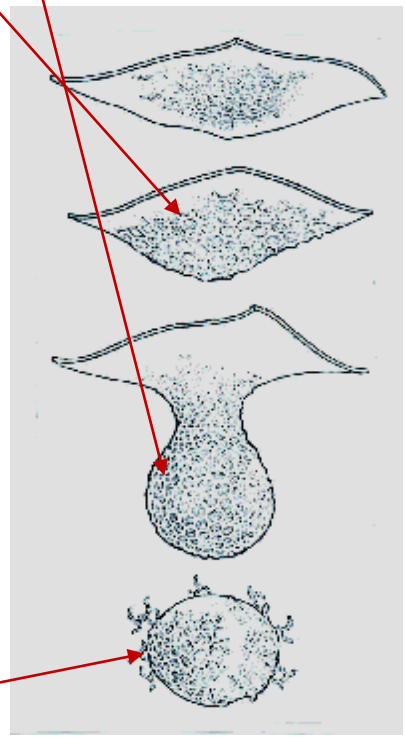
4. **Dynamin** (GTPase activity) strangles membrane to release endocytic vesicle

5. HSP70 (ATPase activity) releases adaptin and clathrin coating to be recycled

RECEPTOR-MEDIATED PINOCYTOSIS (RME)

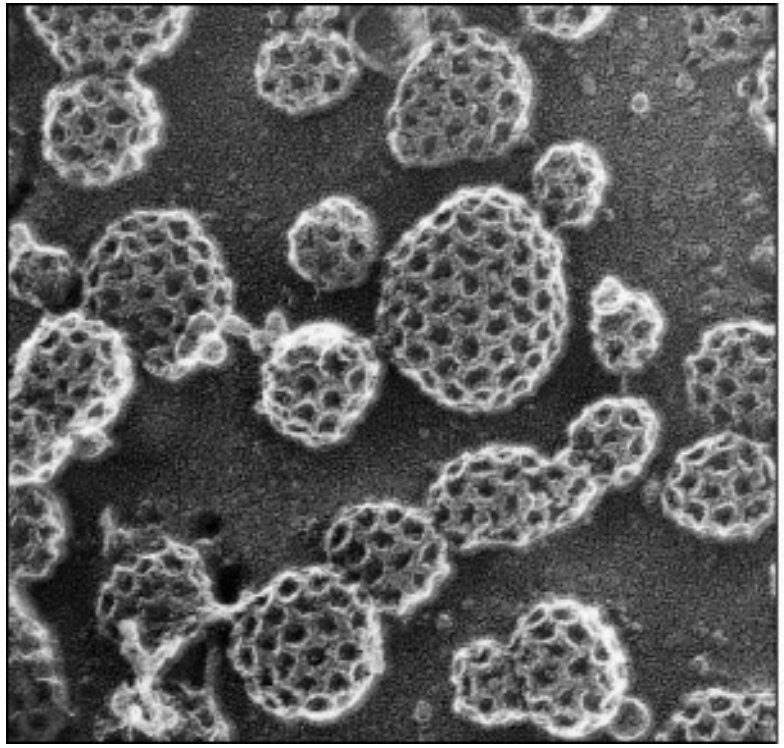
Coated tips





Coated vesicles

Electron microscopy



Freeze-fracture deep etching

RECEPTOR-MEDIATED PINOCYTOSIS (RME)

B. CLATHRIN STRUCTURE

TRISKELION (structural unit)

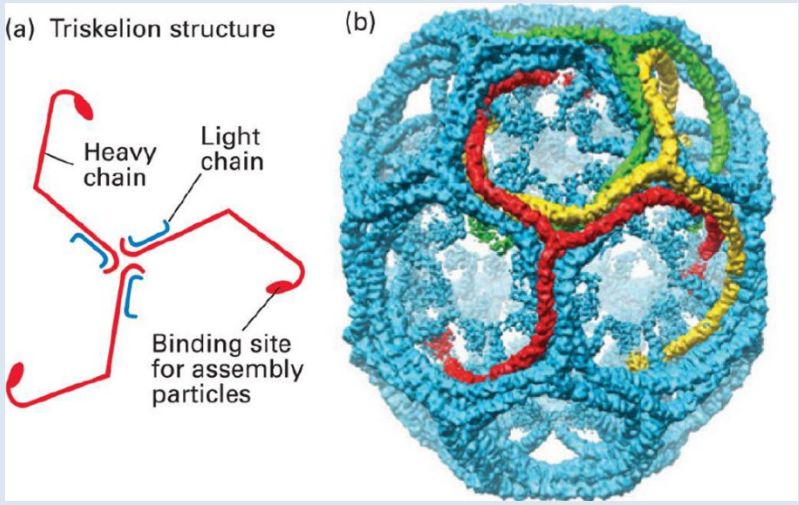
Components

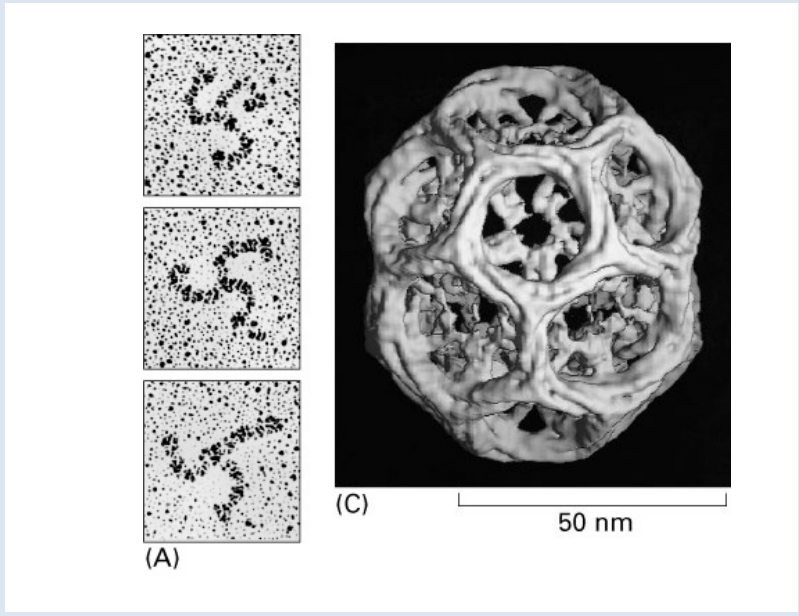
3 heavy chain and

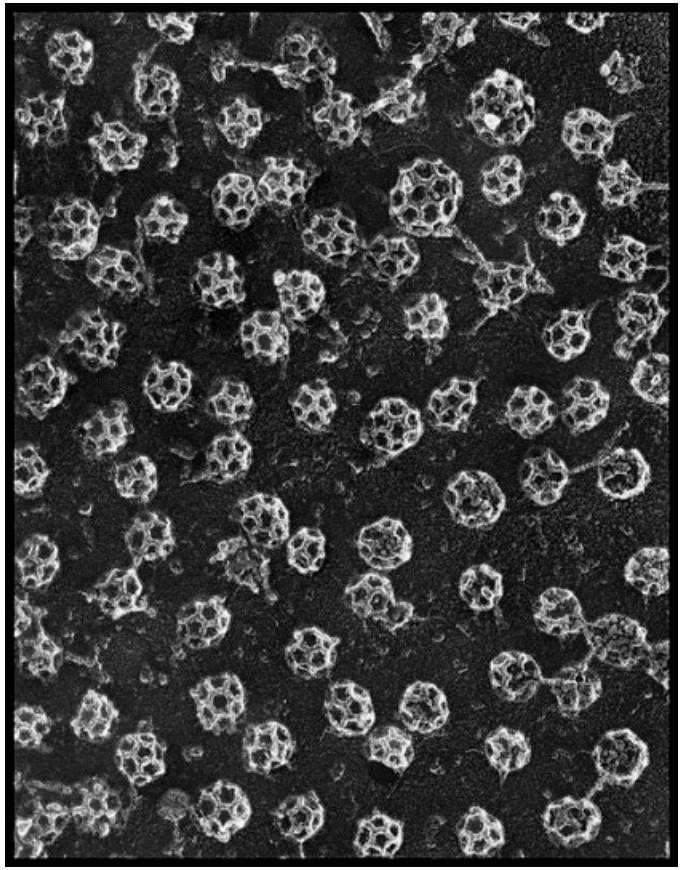
3 light chain peptides

“Honeycomb” or “football” structure

- Formed by 36 triskelions
- 12 pentagons and 8 hexagons







RECEPTOR-MEDIATED PINOCYTOSIS (RME)

C. ENDOSOMAL SORTING

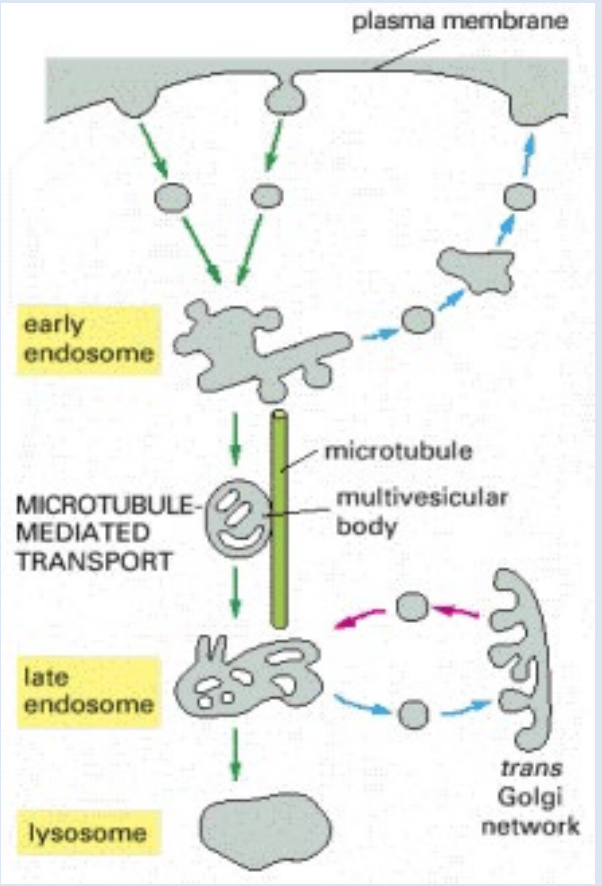
Endosomes: organelles receiving vesicles from plasma membrane or Golgi apparatus

EARLY ENDOSOMES

- **Receive vesicles from plasma membrane**
- Irregular shape
- 1 min to form
- Membrane ATPase decreases pH by pumping in H^+ (pH=6)
- Receptor recycling

LATE ENDOSOMES

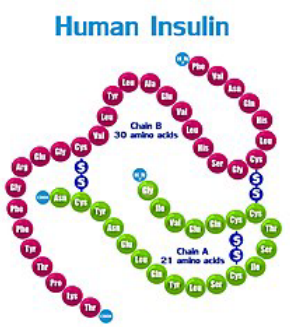
- **Receive vesicles from Golgi apparatus** (to form 1^{ary} lysosomes)
- 15 min to form
- Progressive decrease of pH
- Hydrolases: enzymes degrading of endosome content require low, acidic pH (4.5–5.0)
- Receptor degradation



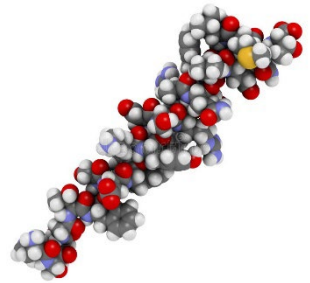
RECEPTOR-MEDIATED PINOCYTOSIS (RME)

D. ENDOCYTOSED MOLECULES

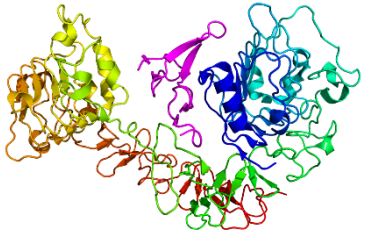
- Hormones (insulin, glucagon)
- Growth factor receptors (e.g., EGFR)
- Lipoproteins (low density lipoproteins)
- Others molecules (e.g., transferrin)
- Viruses and toxins



Glucagon

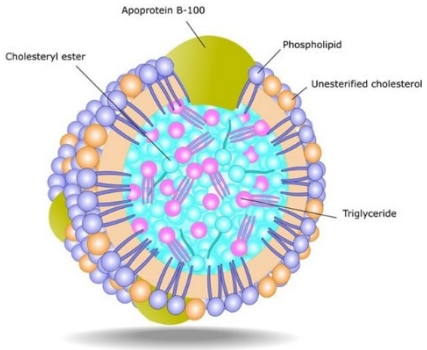


EGFR





LDL



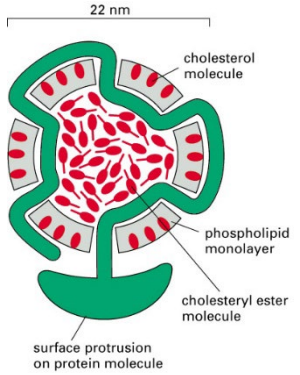
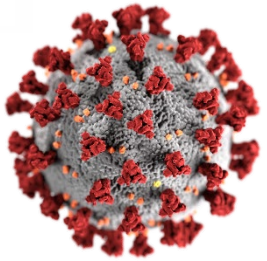


Figure 13–43. Molecular Biology of the Cell, 4th Edition.



Virus

RECEPTOR-MEDIATED PINOCYTOSIS (RME)

E. RECEPTOR FATE

PATHWAY A: RECYCLING

Early Endosome (pH ~6)

Acidic luminal pH dissociates ligand from receptor. Receptors return to cell membrane via recycling endosomes.

Classic Examples:

- Low-Density Lipoprotein Receptor (LDL-R)
- Transferrin Receptor (Iron Delivery)

PATHWAY B: DEGRADATION

Lysosome (pH 4.5–5.0)

Ubiquitinated receptors internalize into multivesicular bodies (MVBs) and fuse with lysosomes for proteolytic breakdown.

Classic Example:

- Epidermal Growth Factor Receptor (EGFR Downregulation)

RECEPTOR-MEDIATED PINOCYTOSIS (RME)

E. RECEPTOR FATE: RECYCLING OF LDL RECEPTOR (1)

Low density lipoproteins (LDL)

Consist of cholesterol, phospholipids and apolipoprotein B

Bad cholesterol (from liver to body cells)

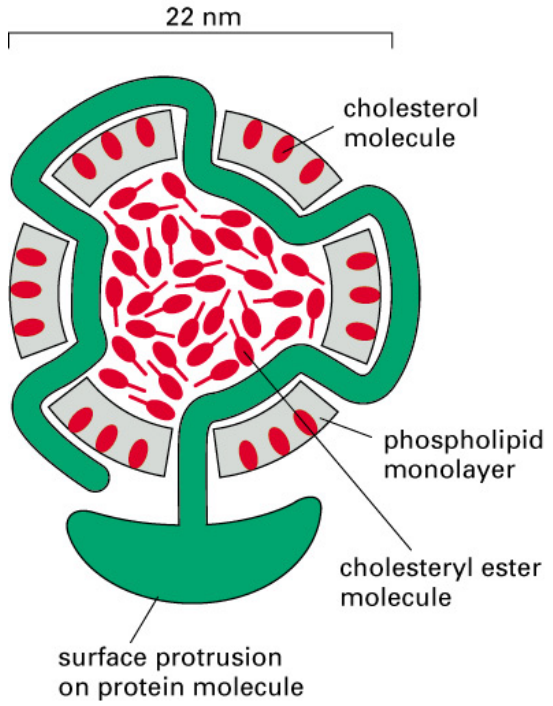
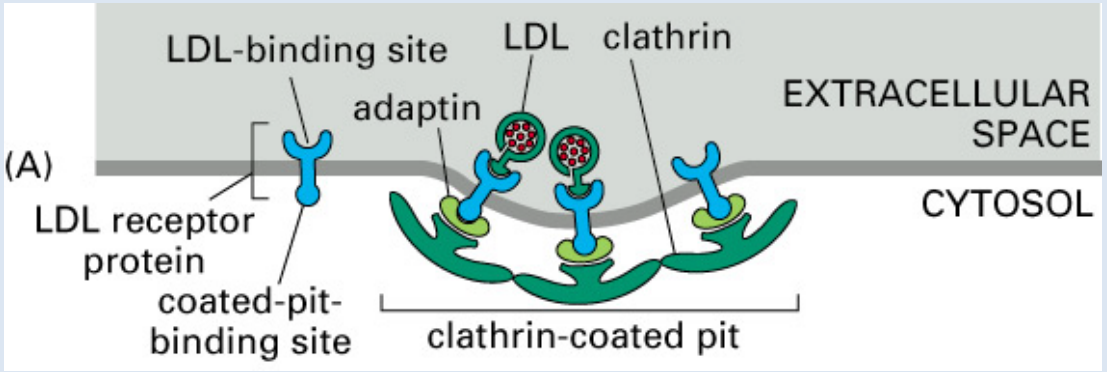


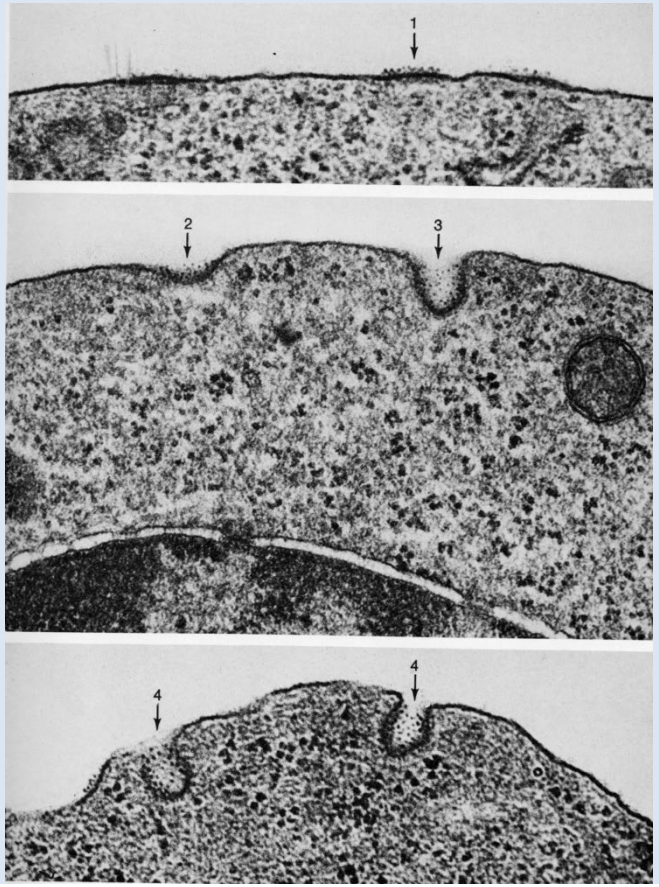
Figure 13–43. Molecular Biology of the Cell, 4th Edition.

RECEPTOR-MEDIATED PINOCYTOSIS (RME)

E. RECEPTOR FATE: RECYCLING OF LDL RECEPTOR (1)

1. Ligand-receptor binding and formation of vesicle

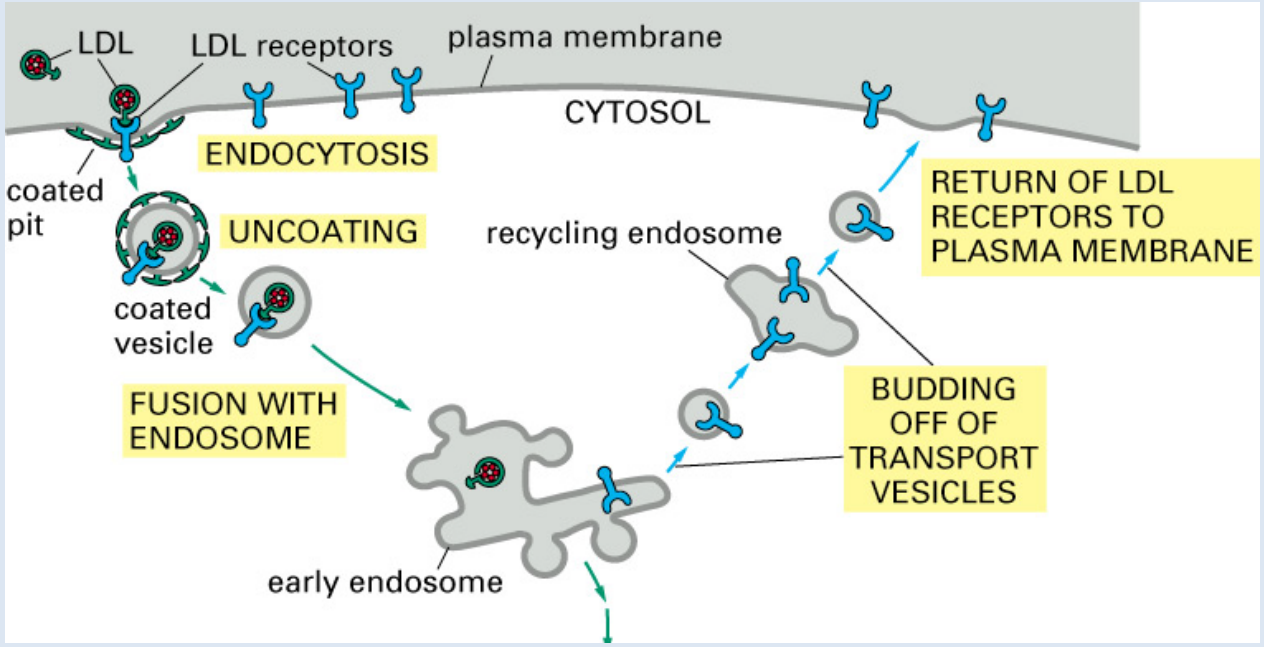




RECEPTOR-MEDIATED PINOCYTOSIS (RME)

E. RECEPTOR FATE: RECYCLING OF LDL RECEPTOR (1)

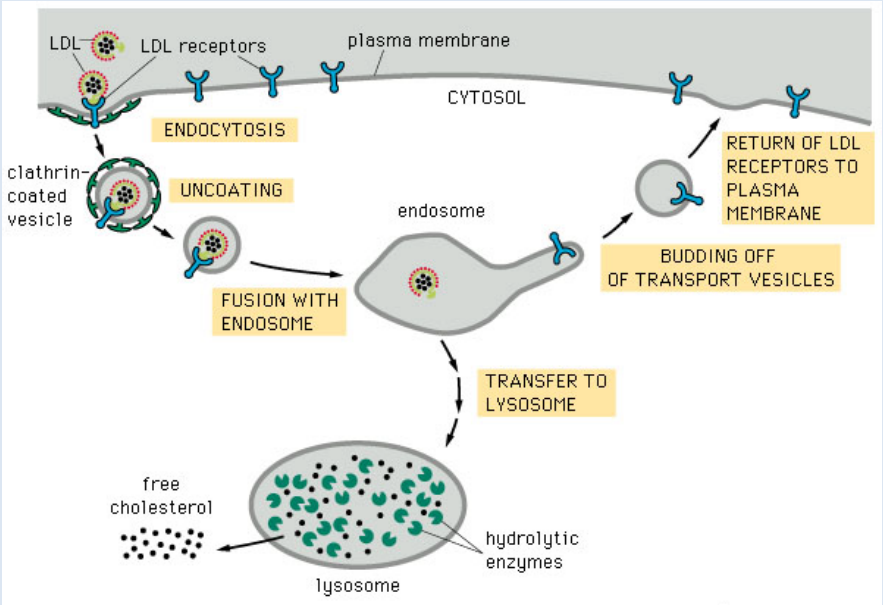
2. Early endosome formation



RECEPTOR-MEDIATED PINOCYTOSIS (RME)

E. RECEPTOR FATE: RECYCLING OF LDL RECEPTOR (1)

3. Endolysosome formation, LDL hydrolization, cholesterol release to cytosol and receptor is recycled through recycling endosomes



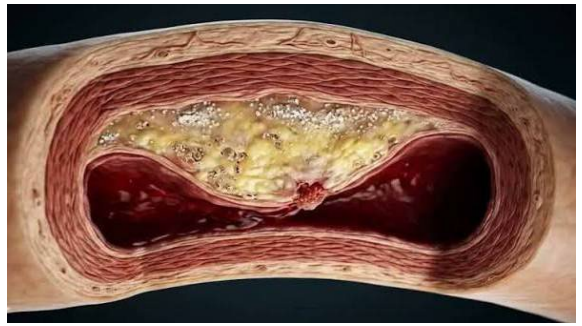
RECEPTOR-MEDIATED PINOCYTOSIS (RME)

E. RECEPTOR FATE: RECYCLING OF LDL RECEPTOR (1)

Abnormal levels of cholesterol

Familial hypercholesterolemia
Mutations in LDL receptor
Affects individuals from infancy
Atherosclerosis / Xanthomas and xanthelasmas

Hypercholesteronemia (adults)
High blood content of colesterol
Diet and life style-related
Atherosclerosis



Atherosclerosis



Xantomas



Xanthelasmas

https://www.youtube.com/watch?v=0U7YHRW5dyc

RECEPTOR-MEDIATED PINOCYTOSIS (RME)

E. RECEPTOR FATE: RECYCLING OF TRANSFERRIN AND TFR (2)

1. **Binding**: Iron-loaded transferrin binds to its specific transferrin receptor (TfR) on the cell membrane.

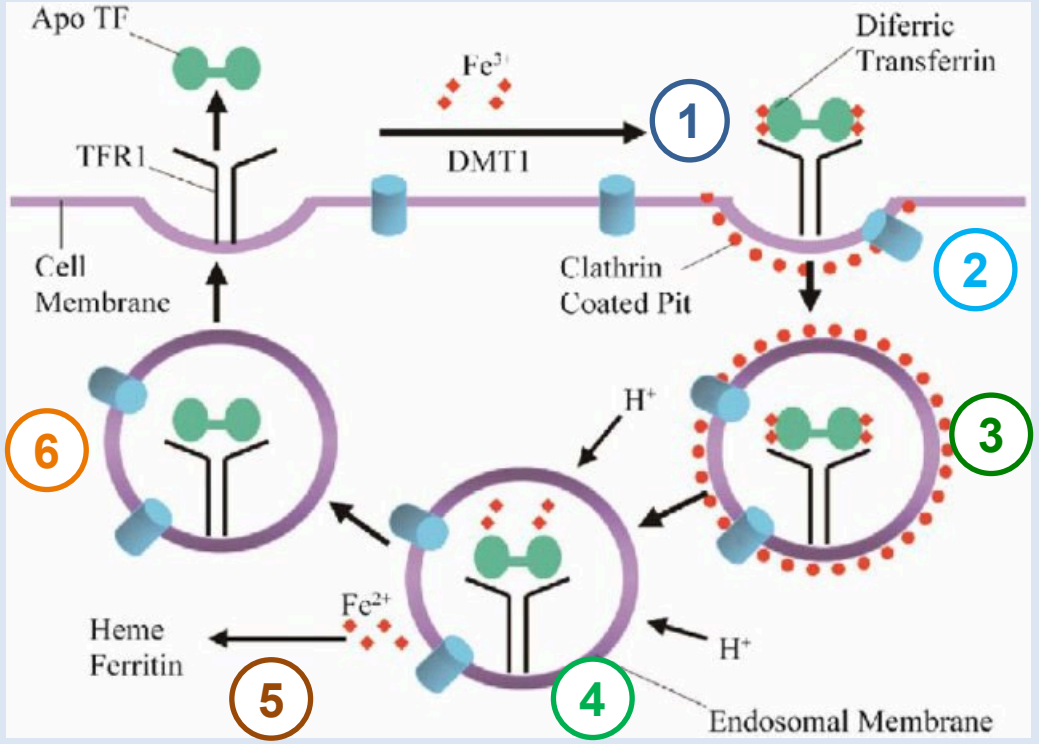
2. **Clathrin-mediated endocytosis**: The binding event stimulates the formation of clathrin-coated pits which include the iron transporter DMT1 (divalent metal transporter 1).

3. **Vesicle formation** (containing the transferrin-TfR complex).

4. **Intracellular trafficking**: The vesicles move into the cytoplasm and deliver their contents to early endosomes.

5. **Iron release** (in the acidic environment of the early endosome).

6. **Recycling**: the iron-free transferrin (apotransferrin) and the TfR are then trafficked to recycling endosomes and back to the cell mbr.



RECEPTOR-MEDIATED PINOCYTOSIS (RME)

E. RECEPTOR FATE: DEGRADATION OF EGFR (3)

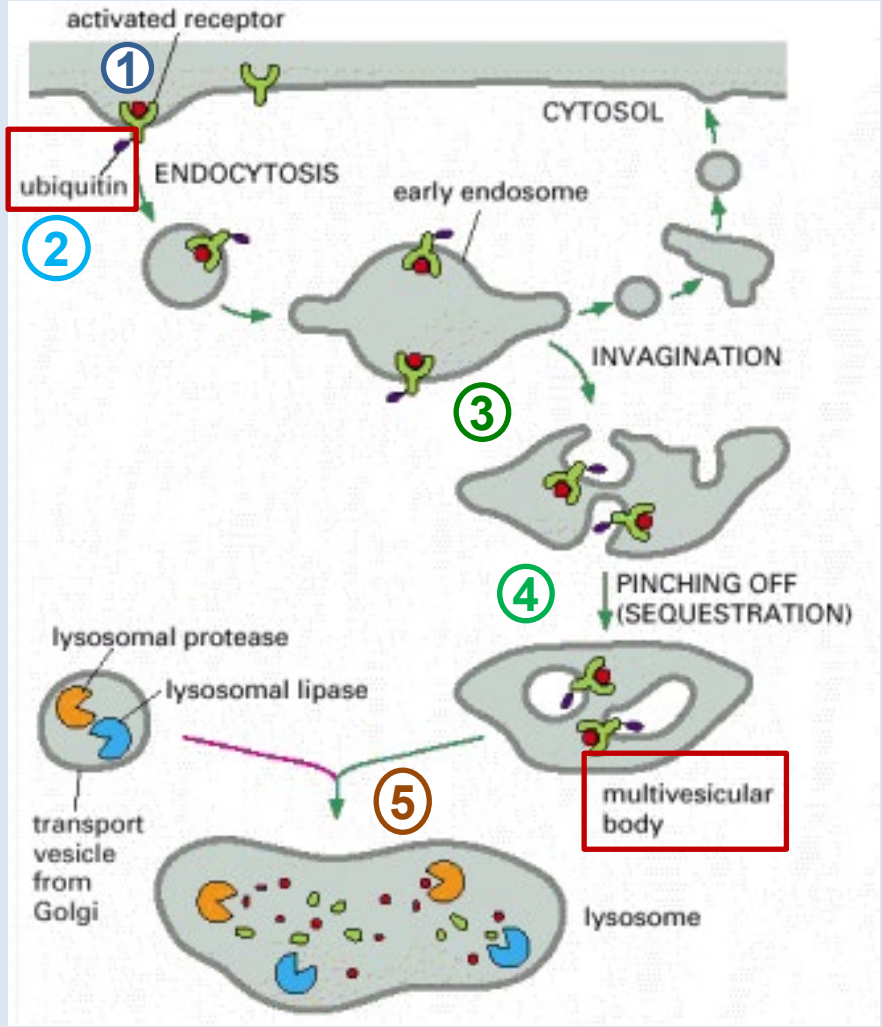
1. **EGF binds EGFR** (cytoplasmic portion triggers a signalling cascade –i.e., ON state-)

2. **Ubiquitination**: Receptor is targeted for degradation to stop the signalling process

3. **No release of ligand-receptor** in early endosomes

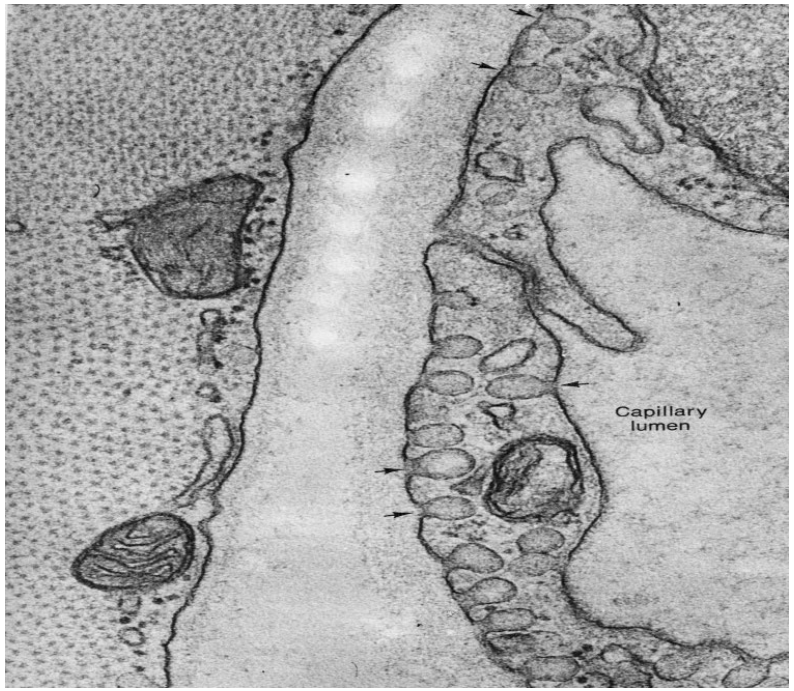
4. **Internalization of cytoplasmic EGFR** in early endosomes and formation of multivesicular bodies

5. **Hydrolization** of EGF-EGFR complexes.



TRANSCYTOSIS

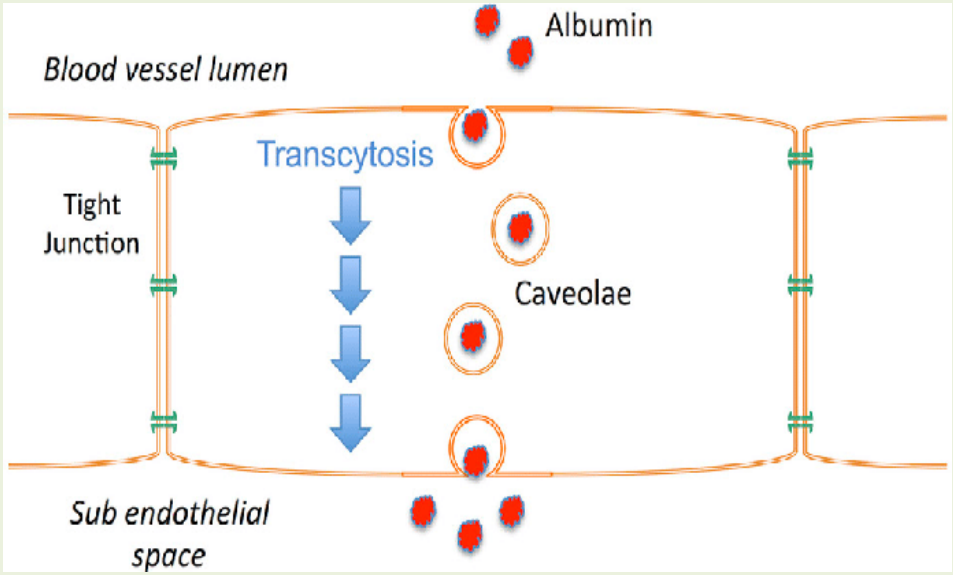
Transportation of a molecule (i.e., ligand) from one part of the cell to another

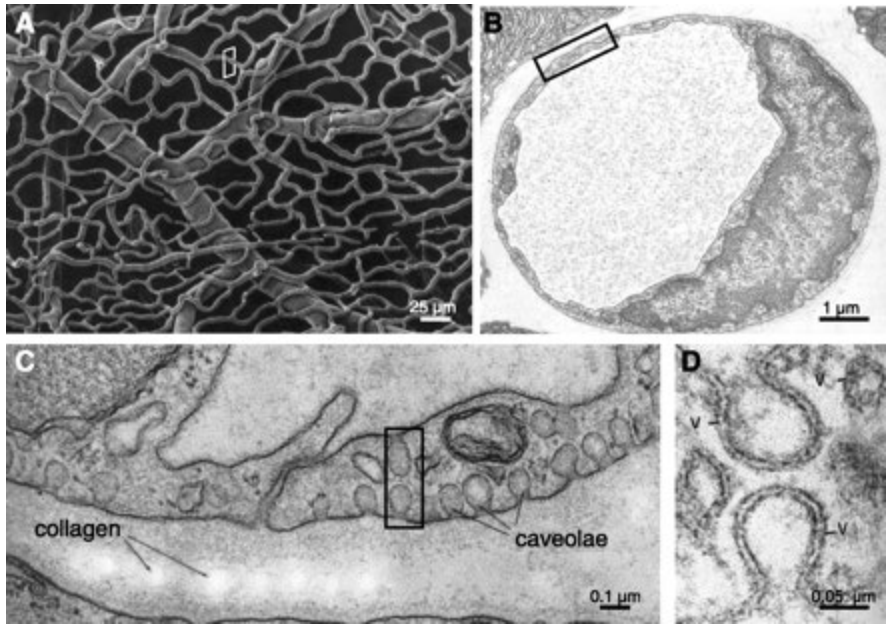


1. NON-RME TRANSCYTOSIS

EXAMPLE: Endothelial cells (capillaries)

- Molecules from blood to tissues or tissues to blood
- **Caveolins**: form caveolae (vesicles)
- **Cavins** control caveola formation/maintenance
- Content: albumin and dextran





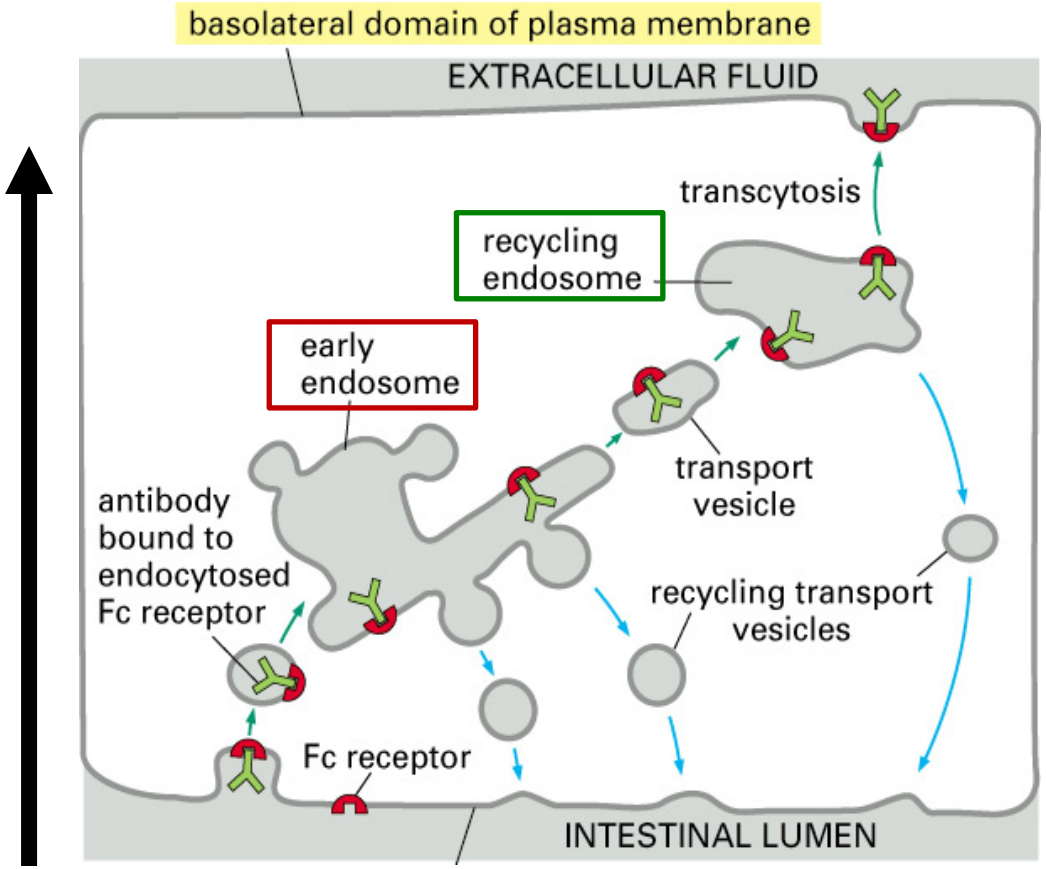
TRANSCYTOSIS

Transportation of a molecule (i.e., ligand) from one part of the cell to another

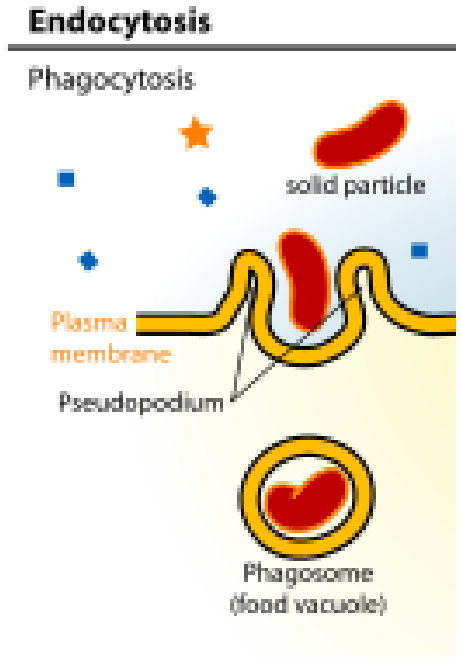
2. RME TRANSCYTOSIS

EXAMPLE: Maternal IgA immunoglobulins to newborns

1. IgAs in maternal milk reach gut epithelium
2. Binding to apical Fc receptors due to acidic pH
3. IgAs get internalized into vesicles
4. Fc recycled to basolateral domain of membrane
5. IgAs are released due to neutral pH



PHAGOCYTOSIS



Swallowing of large particles and micro-organisms

Electron dense content

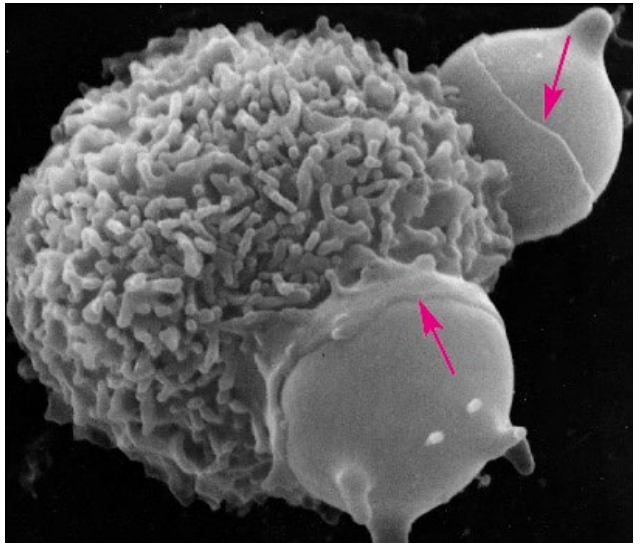
Vacuoles ≥ 0.25 µm (irregular)

Mammals and higher organisms

- Specialized cells or ''professional'' phagocytic cells: macrophages, monocytes, neutrophils, eosinophils, dendritic cells
- Targets: aged, senescent, or apoptotic cells
  hazardous particles
  micro-organisms
- Specialized targeting tissues: spleen-liver (10^{11} aged cells in / day)

Protozoa

Regular way of feeding (macromolecules/bacteria)



PHAGOCYTOSIS

1. Target recognition and adhesion

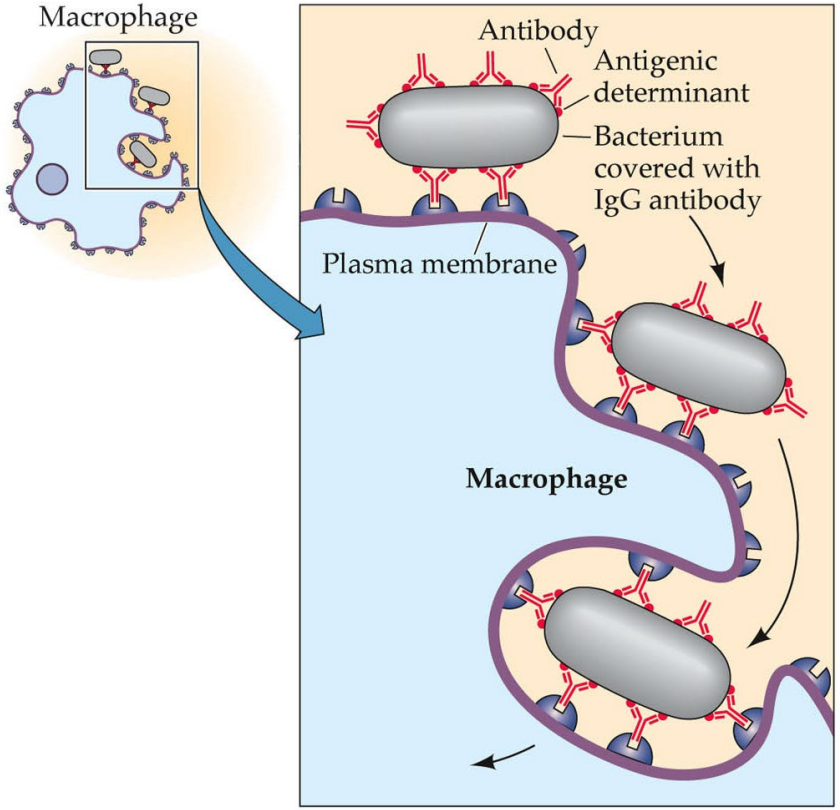
Opsonization (e.g. via antibodies): a pathogen is marked for ingestion-destruction by a phagocyte

1. Antibody produced by immune system
2. Antibody Fab (fragment antigen-binding region) binds pathogen
3. Phagocytic cells bind antibody Fc (crystallizable fraction)

Particles not bound to Ab (inert particles: coal, pollutants…)

2. Target penetration: seudopodes

3. Phagosome and phagolysosome formation



PHAGOCYTOSIS

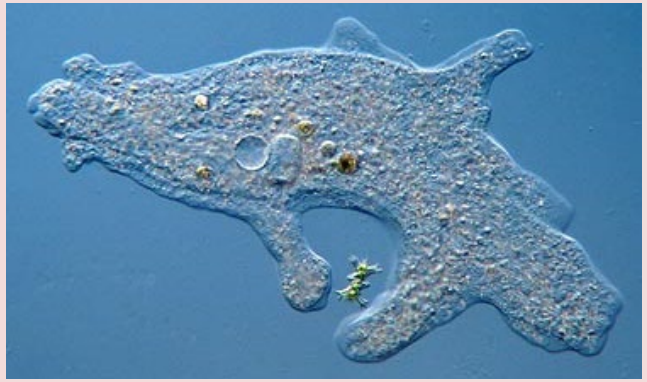
1. Target recognition and adhesion

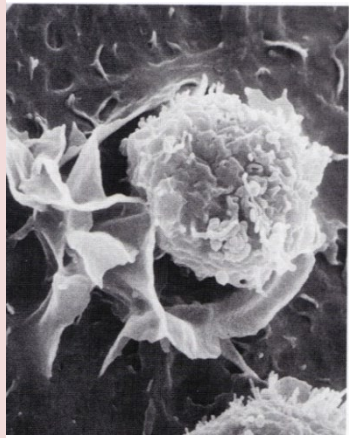
Opsonization (e.g. via antibodies): a pathogen is marked for ingestion-destruction by a phagocyte

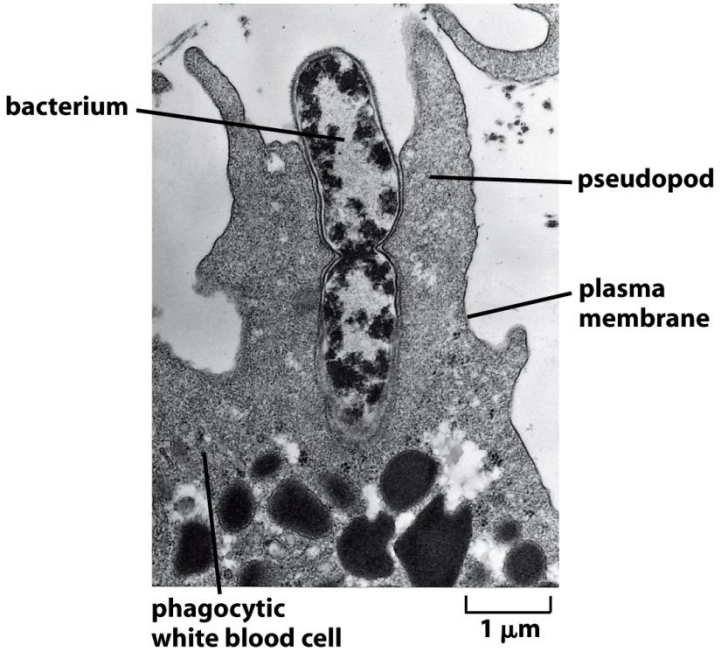
1. Antibody produced by immune system
2. Antibody Fab fragment antigen-binding region) binds pathogen
3. Phagocytic cells bind antibody Fc (crystallizable fraction)

Particles not bound to Ab (inert particles)

2. Target penetration: pseudopodes







PHAGOCYTOSIS

1. Target recognition and adhesion

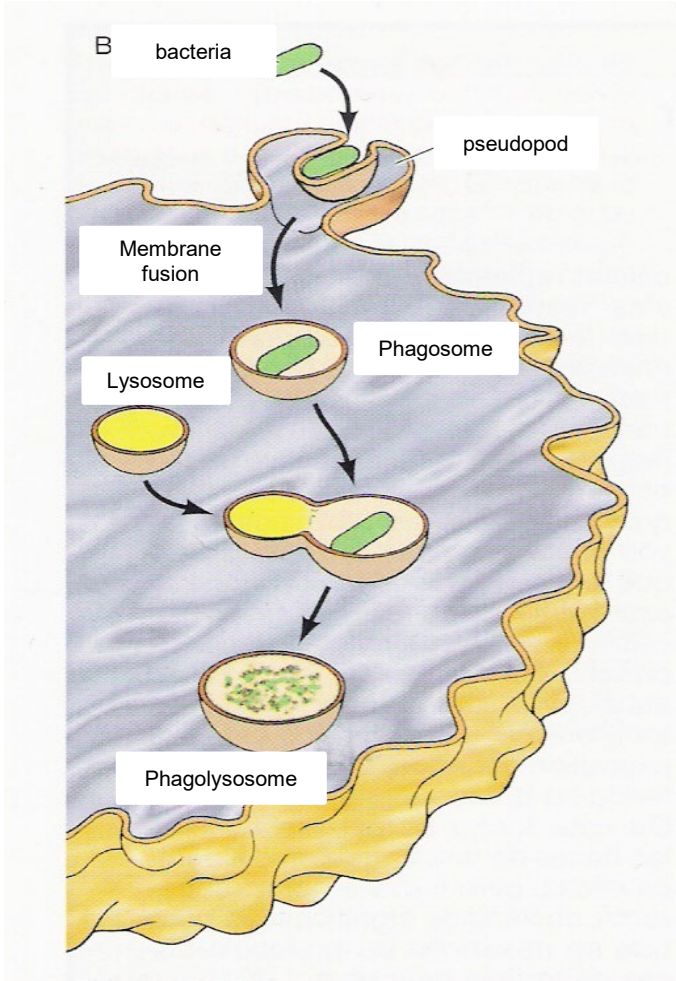
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Particles not bound to Ab (inert particles)

2. Target penetration: seudopodes

3. Phagosome and phagolysosome formation



2. EXOCYTOSIS

Intracellular material transfered to the extracellular space

Exocytosed material

- <u>Disposable material</u>

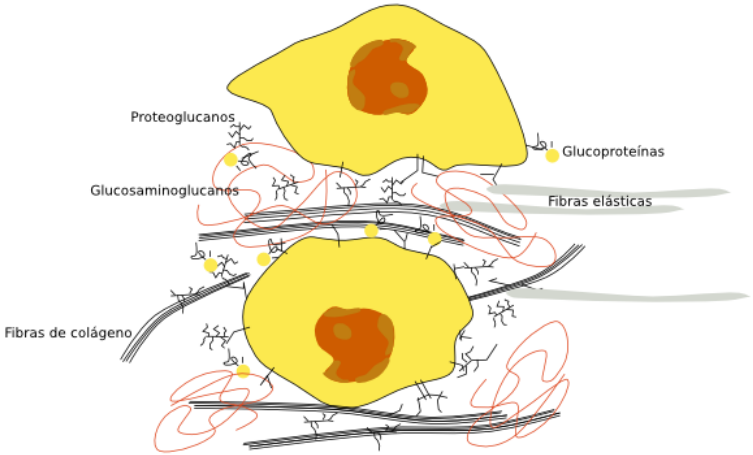
  Products of cell digestion and metabolism

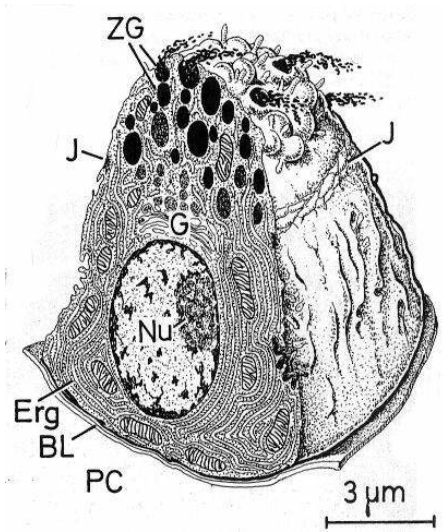
- <u>Cell-synthesized material</u>

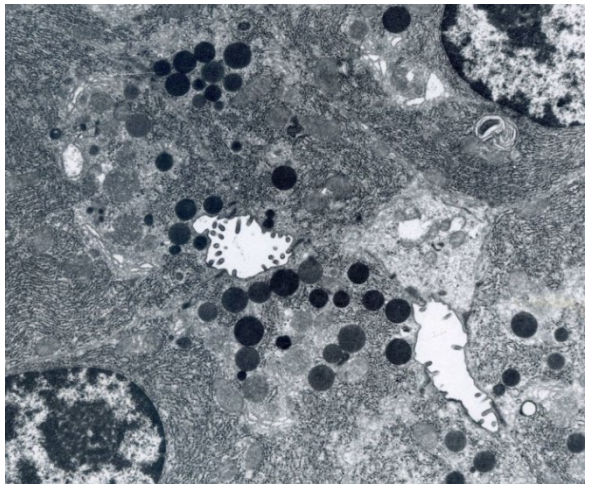
  Cell membrane components (lipids and proteins) and extracellular matrix (all cells)

  Enzymes and mucins (exocrine secretory cells)

  Hormones and signaling molecules (endocrine cells and neurons)







TYPES OF EXOCYTOSIS

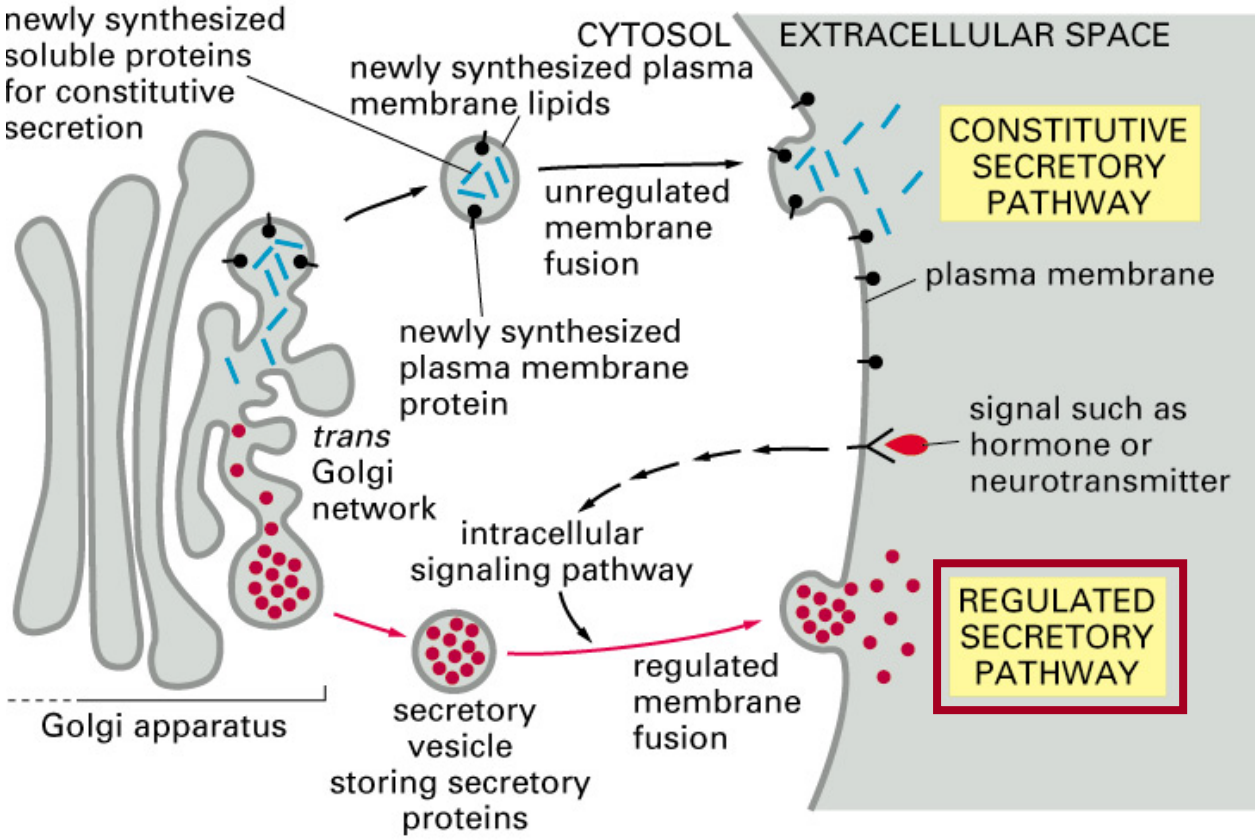


Figure 13–54. Molecular Biology of the Cell, 4th Edition.

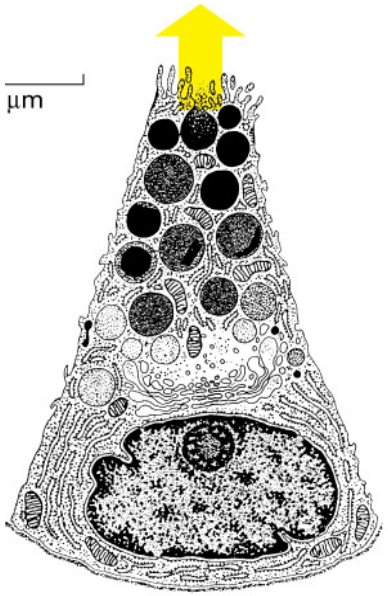
Constitutive

Unregulated – Continuous
All cells
Cargo: Membr. lipids/proteins
Extracelular matrix component

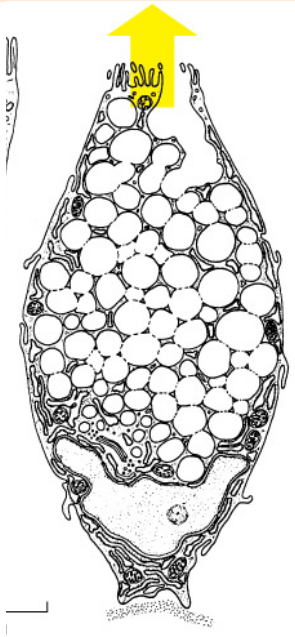
Regulated

External signal or calcium influx
Secretory / stimulated cells
Cargo: granules accumulation in storage vesicles

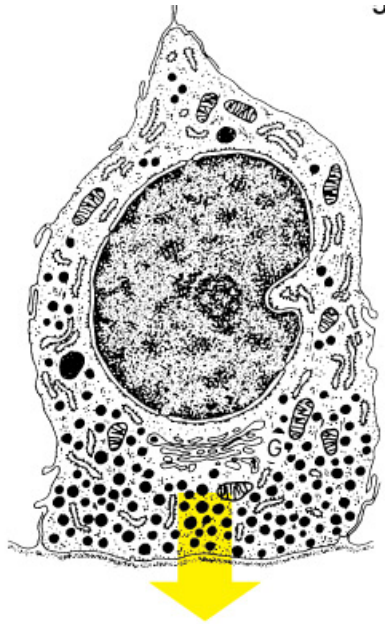
EXAMPLES OF REGULATORY EXOCYTOSIS (SECRETORY CELLS)



Pancreatic acinar cells (prodigestive enzymes, i.e. zymogen granules)



Globet cells (mucin granules)



Endocrine cells (hormones, i.e. insulin and glucagon)